

Oracle AI Database Free Install and Local Database Setup

Windows-focused guide for installing Oracle's free database edition and configuring this Spring Boot project to use a local Oracle database.

Free Edition And Current Version

As of June 17, 2026, Oracle's current free local database page is **Oracle AI Database 26ai Free**.

Oracle lists free resource limits of up to 2 CPU threads, 2 GB of database RAM, and 12 GB of user data.

Oracle also keeps the older Oracle Database XE page, which describes XE as free to use, embed, and redistribute.

Official page: <https://www.oracle.com/database/free/>

Download And Install On Windows

1. Go to <https://www.oracle.com/database/free/>.
2. Under Oracle AI Database Free for Windows, choose Download for Windows.
3. Sign in with an Oracle account if Oracle requires it.
4. Download the Windows x64 installer ZIP or executable for Oracle AI Database 26ai Free.
5. Extract the ZIP if needed, then run setup.exe as Administrator.
6. Choose the install location, usually under C:\app or C:\Program Files\Oracle.
7. Set and save the administrative password for SYS, SYSTEM, and PDBADMIN when prompted.
8. Keep the default listener port 1521 unless another Oracle listener already uses it.
9. Finish the installer and let it create the Windows services.
10. Restart your terminal after installation so PATH changes are available.

Verify The Install

```
sqlplus -v  
  
lsnrctl status
```

For local app development, connect to the pluggable database service. The common default service name for Oracle Database Free is FREEPDB1.

```
sqlplus system@localhost:1521/FREEPDB1
```

Create Local Schema User

Log in with the SYSTEM user, then run:

```
CREATE USER immiauto_user  
  
  IDENTIFIED BY "ChangeThisStrongPassword123!"  
  
  DEFAULT TABLESPACE users  
  
  QUOTA UNLIMITED ON users;  
  
  
GRANT CREATE SESSION TO immiauto_user;  
  
GRANT CREATE TABLE TO immiauto_user;  
  
GRANT CREATE VIEW TO immiauto_user;  
  
GRANT CREATE SEQUENCE TO immiauto_user;  
  
GRANT CREATE PROCEDURE TO immiauto_user;  
  
GRANT CREATE TRIGGER TO immiauto_user;
```

Test the app user:

```
sqlplus immiauto_user/ChangeThisStrongPassword123!@localhost:1521/FREEPDB1
```

Spring Boot Project Configuration

Add the Oracle JDBC driver to backend/pom.xml:

```
<dependency>
    <groupId>com.oracle.database.jdbc</groupId>
    <artifactId>ojdbc11</artifactId>
    <scope>runtime</scope>
</dependency>
```

Use these application.properties database settings:

```
spring.datasource.url=jdbc:oracle:thin:@localhost:1521/FREEPDB1
spring.datasource.driver-class-name=oracle.jdbc.OracleDriver
spring.datasource.username=immiauto_user
spring.datasource.password=ChangeThisStrongPassword123!

spring.jpa.hibernate.ddl-auto=update
spring.jpa.show-sql=true
spring.jpa.properties.hibernate.format_sql=true
spring.jpa.properties.hibernate.dialect=org.hibernate.dialect.OracleDialect

spring.h2.console.enabled=false
```

Run the backend:

```
cd C:\Users\mh200\Projects\Immigration-Consultation\backend
.\mvnw spring-boot:run
```

Connection Summary

```
Host: localhost  
Port: 1521  
Service name: FREEPDB1  
Username: immiauto_user  
Password: ChangeThisStrongPassword123!  
JDBC URL: jdbc:oracle:thin:@localhost:1521/FREEPDB1
```

Use the dedicated immiauto_user for the application. Keep SYS, SYSTEM, and PDBADMIN only for administration.

Official References

Oracle AI Database Free: <https://www.oracle.com/database/free/>

Oracle Database XE: <https://www.oracle.com/database/technologies/appdev/xe.html>